

Unequal Job Opportunities and Well-Being Inequality: A Latent-Jobs Structural Decomposition

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France: EU-SILC collected in 2016, incomes of 2015, every job priced through EUROMOD.

Discussion draft. The decomposition is preliminary and is not a causal decomposition.

Two households with the same income can be very differently placed.

- Low earnings can mean a taste for short hours, no well-paid job within reach, or household resources that make another arrangement affordable.
- An income distribution records outcomes with these explanations already mixed in.
- In France, well-being inequality differs from consumption inequality:
 - single adults: consumption Gini 0.2633, attained-bundle well-being Gini 0.2498;
 - couples: consumption Gini 0.2268, attained-bundle well-being Gini 0.1974.

Equivalised, household-weighted, within each population. Single-adult and couple households are separate populations and are not compared with each other.

Each ingredient exists; a decomposition of the level of well-being inequality across them does not.

Labour supply as choice among latent jobs

Aaberge, Dagsvik and Strøm;
Dagsvik and Strøm; Dagsvik and Jia; Capéau, Decoster and Dekkers.

Separates tastes from opportunities; used to evaluate *reforms*.

Money-metric welfare with heterogeneous preferences

Fleurbaey and Maniquet; Decoster and Haan; Bargain et al.; Jia and Thoresen; Jacquet, Jia and Thoresen.

A reference encodes a normative position; references sit on deterministic budget sets, or value a *change* between regimes.

Decomposing inequality

Shorrocks; Sastre and Trannoy; Creedy and Hérault; Mühlhan.

Shapley allocation of a *change* between policy, population or time states.

Gap: a cross-sectional *level* of well-being inequality, decomposed into preferences and job opportunities inside an estimated latent-jobs model, under welfare measures that can value opportunities themselves.

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?
2. How do conclusions differ when welfare evaluates the attained bundle versus the ex-ante opportunity prospect?
3. In a restricted P/A/B decomposition, how much welfare inequality is associated with preferences, local access, and earning opportunities?

The latent-jobs model: a job is a package, valued through leisure and consumption.

A package j : employment, one of four occupation groups, weekly hours and an hourly wage, priced through the French tax-benefit system to monthly disposable consumption $C_i(j)$.

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right)$$

L_i value of leisure, shifted by age and, for women, children β_c weight on log consumption, estimated λ_c a units convention

Estimated separately for 1,540 single-adult and 2,223 couple households. The child term is a reduced-form time-constraint shifter, not pure taste.

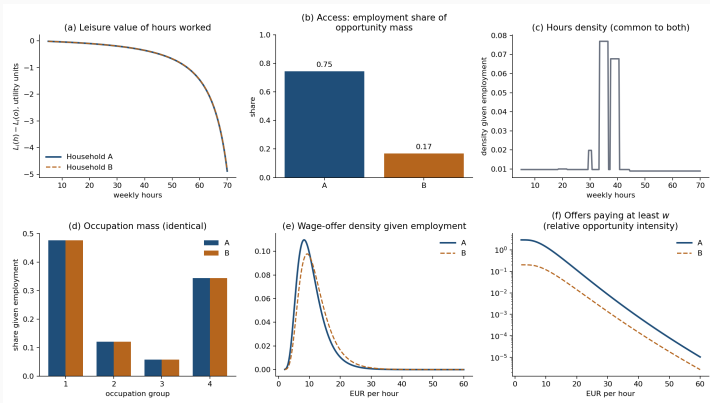
A job is chosen because it is valued or because it is plentiful: the choice law carries an opportunity density.

$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$

g_i : how intensely package j is available to household i before any choice. It has four blocks: **access** (local unemployment exposure, region, urban or rural location, year), hours bands, occupation, and **wage offers** (education, experience).

Separation rests on maintained restrictions: smooth preferences against banded hours opportunities; local shifters excluded from preferences; wage offers independent of hours given occupation.

What unequal opportunity means here: two matched men with almost identical tastes.



Same occupation group, hours band and wage quintile. A has 14.5 times B's employment mass; B's wage-offer location is 11.5 log points higher. A stated-rule teaching example: not causal, not representative.

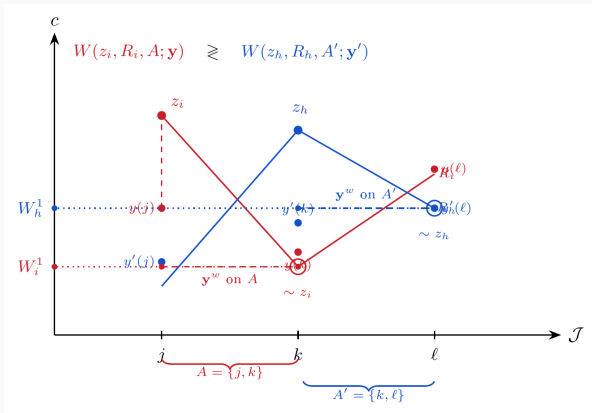
First subquestion: choices reflect opportunities as well as preferences.

Single adults	free coordinates	population fit (mean absolute deviation)
Latent jobs, household-specific opportunities	41	0.0140
Common opportunity distribution	10	0.0273
Common opportunity shape; employment and hours moved into utility	16	0.0264

Both re-estimated common-opportunity benchmarks are worse by at least 141.64 on the same criterion, households and sampled alternatives; the occupation margins deteriorate by an order of magnitude.

Not converted into a formal test. A better criterion does not by itself establish that a mechanism has been identified.

From choices to well-being: give every job the same consumption and ask what is as good as the actual situation.



Own-set equal-consumption equivalents: the theoretical construction from the companion theory paper. Two individuals, with preferences R_i and R_h and ability sets $A = \{j, k\}$ and $A' = \{k, \ell\}$, attain bundles z_i and z_h . For each individual, a common consumption level is assigned to every job in their own set; the level at which the preferred reference job becomes indifferent to the attained bundle is that individual's money metric, W_i^1 and W_h^1 . The two metrics are then directly comparable. Adapted from Haydar and Maniquet (2026), work in progress. This is the deterministic construction; the estimated ex-ante measure is its extension, defined in Section 3.

EA, the opportunity-prospect perspective: what flat consumption makes the whole job prospect equally valuable?

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j), \quad H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$

$$M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

- J_i : the actual prospect; every job enters, weighted by how available it is and how much it is valued.
- H_i : the same jobs, availability and preferences, with pay differences removed.
- Opportunities enter welfare directly: access moves which jobs carry weight; earning opportunities move what they pay.

The measure treats taste-shock variety as part of what makes a rich prospect valuable: a normative position, not a consequence of estimation.

ATT, the attained-bundle benchmark: what consumption without work is as good as the job actually held?

$$M_i^{\text{att}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}$$

- Reference: non-employment o , which every household can reach.
- Opportunities matter only through which job is attained and what it pays; jobs not taken play no role.
- For non-workers $M_i^{\text{att}} = C_i^{\text{obs}}$; for workers it lies below observed consumption by the value of the leisure given up.

On the current empirical domain the equal-consumption reference job is non-employment for every household: a verified property of the estimated specification, not a general theorem.

The two measures answer different welfare questions.

	EA: the opportunity prospect	ATT: the attained bundle
Question	How valuable is the distribution of job prospects the household faces?	How well off is the household in the bundle it actually attains?
Reference	The same opportunity environment, with consumption equalised across jobs	A common non-employment state
How opportunities enter	Directly, as weights on every job	Only through the realised job

Neither perspective corrects the other, and neither is designated primary. The choice between them is a substantive normative decision that is not made here.

The decomposition equalises three channels and allocates the change in the Gini over all orders.

P preferences: age profiles and the child-related shifter.

A local access: local unemployment exposure, region, urban or rural location, year. Not total opportunity.

B earning opportunities: systematic wage-offer location, not wage dispersion.

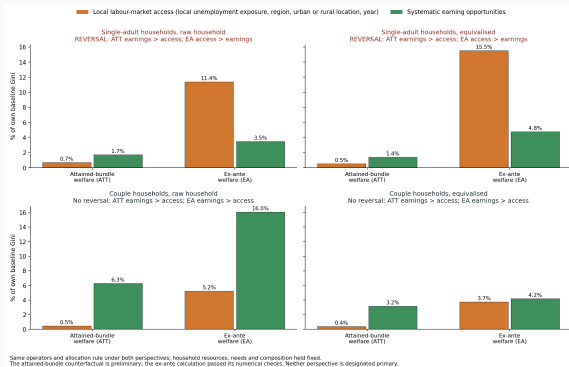
Held fixed: household resources, needs and composition, in every coalition, with no allocated share.

$$\phi_k^p = \sum_{S \subseteq \{P, A, B\} \setminus \{k\}} \frac{|S|! (3 - |S| - 1)!}{3!} \left[v^p(S \cup \{k\}) - v^p(S) \right]$$

$v^p(S) = I^p(\emptyset) - I^p(S)$: the fall in the household-weighted Gini of welfare M^p , $p \in \{EA, att\}$, when the channels in S are set to a common reference.

Household resources, needs and composition are held fixed in the current decomposition. The access channel is local access, not total opportunity. The decomposition is preliminary and makes no causal claim.

Central result: earning opportunities dominate under ATT; access dominates under EA for single adults; couples show no reversal.



Access and earning-opportunity contributions as a percentage of each perspective's own baseline Gini. Access is local access, not total opportunity; household resources, needs and composition are held fixed. Neither perspective is designated primary.

EA: access and earnings account for 7.9–21.3% of baseline inequality; for single adults access is about three times earnings.

	scale	access Gini points	earnings Gini points	access + earnings % of baseline	ordering
Single adults	raw	0.01812	0.00555	14.8	access > earnings
	equivalised	0.02183	0.00673	20.3	access > earnings
Couples	raw	0.00667	0.02049	21.3	earnings > access
	equivalised	0.00449	0.00504	7.9	earnings > access

Single adults: access is 3.3 times earnings before equalisation and 3.2 times after it.

Household resources, needs and composition held fixed; access is local access. For couples, equalisation moves the share from 21.3% to 7.9% and turns the preference term negative, so no directional claim about preferences is made.

ATT benchmark: access and earnings account for 1.9–6.7% of baseline inequality, and earning opportunities are larger throughout.

	scale	access Gini points	earnings Gini points	access + earnings % of baseline	ordering
Single adults	raw	0.0016	0.0040	2.4	earnings > access
	equivalised	0.0013	0.0034	1.9	earnings > access
Couples	raw	0.0009	0.0128	6.7	earnings > access
	equivalised	0.0007	0.0062	3.5	earnings > access

Preliminary: the attained-bundle counterfactual integration is still being validated and covers short hours sparsely. The preference contribution changes sign with equivalisation in both samples, so no directional claim is made about it. Household resources, needs and composition held fixed.

What these results are not.

- **Preliminary.** A restricted structural accounting exercise; the attained-bundle counterfactual integration is still being validated.
- **No causal claim.** Separation rests on maintained functional-form and exclusion restrictions; no causal effect of geography, education or occupation is estimated.
- **No parameter uncertainty yet.** Estimation uncertainty is not propagated through either decomposition; repeat runs are simulation checks, not confidence intervals.
- **Preferences are not responsibility.** The preference channel is systematic utility heterogeneity, including a reduced-form child-related shifter; it is not equated with what households are responsible for.
- **Held fixed and narrow.** Household resources, needs and composition are held fixed; access is local access, not total opportunity. The shares are not the total share of inequality due to unequal job opportunities.

Conclusion: which labour-market inequality matters depends on the welfare question.

1. **Choices reflect opportunities, not preferences alone.** Common-opportunity benchmarks fit worse on the same households.
2. **The welfare question changes the answer.** Under **EA**, access and earnings account for 7.9–21.3% of baseline inequality and access leads for single adults; under **ATT**, 1.9–6.7% and earnings lead. Couples show no reversal.
3. **In the restricted P/A/B decomposition**, local access and earning opportunities are a bounded part of well-being inequality; the preference contribution is not sign-robust to equivalisation.

Next: household resources, needs and composition as a fourth channel; parameter uncertainty; completing validation of the attained-bundle integration.